The Spectrum of an Operator

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Outline

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Revision

Definition

Let T be a linear operator, $\lambda \in \mathbb{C}$ is called an eigenvalue of T, if there exists $x \in H$ with $x \neq 0$ such that

$$Tx = \lambda x$$

Revision

Theorem

Let $\mathcal{H}$ be a Hilbert space and $T : \mathcal{H} \to \mathcal{H}$ then T is invertible $\Longleftrightarrow$ $\ker T^* = \{0\}$ and $\exists \alpha > 0$ such that $\|T(x)\| \geqslant \alpha \|x\| \quad \forall x \in \mathcal{H}$.

Revision

Theorem

Let $\mathcal{H}$ be a Hilbert space and $T : \mathcal{H} \to \mathcal{H}$ then T is invertible $\iff$ $\ker T^* = \{0\}$ and $\exists \alpha > 0$ such that $\|T(x)\| \geqslant \alpha \|x\| \quad \forall x \in \mathcal{H}$.

Corollary

Let H be a Hilbert space and $T : H \to H$ then T is invertible $\iff$ $\exists \beta > 0$ such that $\|T^*(x)\| \geqslant \beta \|x\|$ and $\exists \alpha > 0$ such that $\|T(x)\| \geqslant \alpha \|x\| \quad \forall x \in H$.

Revision

Proof.

If T is invertible, then so is T^*. So, by Theorem 1, $\exists\beta > 0$ such that $\|T^*(x)\| \geqslant \beta\|x\|$ and $\exists\alpha > 0$ such that $\|T(x)\| \geqslant \alpha\|x\|$ for all $x \in H$. Conversely, if $\exists\beta > 0$ such that $\|T^*(x)\| \geqslant \beta\|x\|$, then if $T^*u = 0$ for some $u \in H$, then

$$0 = \|T^*u\| \geq \beta\|u\|$$
$$\implies u = 0$$

So, $\ker T^* = \{0\}$. By Theorem 1, T is invertible. $\square$

Outline

Definition of Spectrum

Definition

Let T be a linear operator. The resolvent set of T, denoted by $\rho(T)$, is $\rho(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is invertible}\}$. The spectrum of T, denoted by $\sigma(T)$ is $\sigma(T) = \mathbb{C} \backslash \rho(T)$, or equivalently, $\sigma(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is not invertible}\}$

Definition of Spectrum

Remark

Notice that, if $T \in B(\mathcal{H})$, by Theorem 1, we can rewrite the resolvent set of T as

$$\rho(T) = \{\lambda \in \mathbb{C} : \ker(T - \lambda I)^* = \{0\} \\ \text{and } \|(T - \lambda I)x\| \geq \alpha\|x\| \text{ for some } \alpha > 0\}$$

Alternatively, we can rewrite the resolvent set of T as

$$\rho(T) = \{\lambda \in \mathbb{C} : \|(T - \lambda I)^* x\| \geq \beta\|x\| \\ \text{and } \|(T - \lambda I)x\| \geq \alpha\|x\| \text{ for some } \beta, \alpha > 0\}$$

by Corollary 1

Relationship between Eigenvalue and Spectrum

Theorem

If λ is an eigenvalue of T , then $\lambda \in \sigma(T)$

Relationship between Eigenvalue and Spectrum

Proof.

Suppose that $\lambda \notin \sigma(T)$, then $T - \lambda I$ is invertible. So, we have $(T-\lambda I)u = 0 \implies u = (T-\lambda I)^{-1}(T-\lambda I)u = (T-\lambda I)^{-1}(0) = 0$. So, if there is $u \in \mathcal{H}$ such that $Tu = \lambda u$, then $u = 0$. So, λ is not an eignvalue of T. $\square$

Relationship between Eigenvalue and Spectrum

Theorem

Let H be a finite dimensional Hilbert space, and let $T : H \to H$ be linear. If $\lambda \in \sigma(T)$, then λ is an eigenvalue of T.

Relationship between Eigenvalue and Spectrum

Remark

For finite dimesonial Hilbert space, the spectrum of T consist only the eigenvalues of T. However, in infinite dimesional Hilbert space, we may have no eigenvalues. For instance, consider the shift operator. Here, it has no eigenvalues, since if we consider $(T - \lambda I)x = 0$, we see that

$$(-\lambda x_1, x_1 - \lambda x_2, x_2 - \lambda x_3, \cdots) = (0, 0, 0, \cdots) \implies x = (0, 0, 0, \cdots)$$

So, T does not have an eigenvalue. However, we showed that T is not invertible. So, $T - 0I$ is not invertible and $0 \in \sigma(T)$. So, the spectrum of an operator does not only consist of eigenvalues.

Relationship between Eigenvalue and Spectrum

Proof.

Suppose that λ is not an eigenvalue of T, then there is nonzero $u \in H$ such that $Tu = \lambda u$. So, there is no nonzero $u \in H$ such that $(T - \lambda I)u = 0$. So, $\ker(T - \lambda I) = \{0\}$. By rank-nullity theorem, we have

$$\begin{aligned} \dim H &= \dim \ker(T - \lambda I) + \dim \operatorname{Im}(T - \lambda I) \\ &= 0 + \dim \operatorname{Im}(T - \lambda I) \\ &= \dim \operatorname{Im}(T - \lambda I) \end{aligned}$$

So, $\operatorname{Im}(T - \lambda I) = H$. This means that $(T - \lambda I)$ is surjective, and therefore, invertible. So, $\lambda \in \rho(T)$, and therefore, $\lambda \notin \sigma(T)$. $\square$

Properties and Theorems

Proposition

Let $\mathcal{H}$ be a Hilbert space. Let $T : \mathcal{H} \to \mathcal{H}$ be linear operator. The following properties hold:

1. $\sigma(I) = \{1\}$
2. $\sigma(T^*) = \{\bar{\lambda} : \lambda \in \sigma(T)\}$
3. *If T is invertible, then* $\sigma\left(T^{-1}\right) = \left\{\mu^{-1} : \mu \in \sigma(T)\right\}$
4. *If T is self-adjoint, then $\sigma(T) \subseteq \mathbb{R}$.*
5. *If $U : \mathcal{H} \to \mathcal{H}$ is invertible, then $\sigma(UTU^{-1}) = \sigma(T)$*

Properties and Theorems

Lemma

If $\mathcal{H}$ is a complex Hilbert space and $U \in B(\mathcal{H})$ is unitary then

$$\sigma(U) \subseteq \{\lambda \in \mathbb{C} : |\lambda| = 1\}$$

Properties and Theorems

Theorem

Let $\mathcal{H}$ be a complex Hilbert space and let $T \in B(\mathcal{H})$. If $|\lambda| > \|T\|$ then $\lambda \notin \sigma(T)$.

Properties and Theorems

Corollary

If $T \in B(\mathcal{H})$, then $\sigma(T)$ is bounded.

Properties and Theorems

Proof.

If $\lambda \in \sigma(T)$, then, by the contrapositive of Theorem 4, $|\lambda| \leq \|T\|$. So, $\lambda \in \overline{B(0, \|T\|)}$. Hence, $\sigma(T) \subseteq \overline{B(0, \|T\|)}$, i.e. $\sigma(T)$ is bounded. $\square$

Properties and Theorems

Theorem

If $T \in B(\mathcal{H})$, *then* $\sigma(T) \subseteq \mathbb{C}$ *is a closed set.*

Properties and Theorems

Corollary

If $T \in B(\mathcal{H})$, then $\sigma(T)$ is compact in $\mathbb{C}$

Outline

The Shift Operator

We would like to find the spectrum of the unilateral shift, $T : \ell^2 \to \ell^2$, defined by

$$T(x_1, x_2, \cdots) = (0, x_1, x_2, \cdots)$$

Since T has been shown to have no eigenvalue, we will consider the eigenvalues of T^* instead and use the properties of the spectrum to have a lower bound for the spectrum.

The Shift Operator

Suppose that $\lambda \in \mathbb{C}$ such that $|\lambda| < 1$, then consider the sequence

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We have

$$\sum_{n=0}^{\infty} |x_n|^2 = \sum_{n=0}^{\infty} (|\lambda|^n)^2 = \sum_{n=0}^{\infty} (|\lambda|^2)^n = \frac{1}{1 - |\lambda|^2} < \infty$$

The Shift Operator

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$$\begin{aligned} T^*x &= T^*(1, \lambda, \lambda^2, \cdots) \\ &= (\lambda, \lambda^2, \lambda^3, \cdots) = \lambda(1, \lambda, \lambda^2, \cdots) = \lambda x \end{aligned}$$

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So, every $\lambda \in \mathbb{C}$ suhc that $|\lambda| < 1$ is an eigenvalue of T^*. Now, we have

$$\{\lambda : |\lambda| < 1\} = \{\bar{\lambda} : |\lambda| < 1\} \subseteq \sigma(T)$$

The Shift Operator

So,

$$\{\lambda \in \mathbb{C} : |\lambda| < 1\} \subseteq \sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

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So,

$$\{\lambda \in \mathbb{C} : |\lambda| < 1\} \subseteq \sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

or

$$B(0,1) \subseteq \sigma(T) \subseteq \overline{B(0,1)}$$

The Shift Operator

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$$\{\lambda \in \mathbb{C} : |\lambda| < 1\} \subseteq \sigma(T) \subseteq \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

or

$$B(0,1) \subseteq \sigma(T) \subseteq \overline{B(0,1)}$$

Since $\sigma(T)$ is closed, we have

$$\sigma(T) = \overline{B(0,1)} = \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$